TIANMU YUAN

Postdoctoral Research Associate, Newcastle University

Email | Homepage | ORCiD | Linkedin | GitHub | Google Scholar

RESEARCH INTEREST

Research themes: Solid-state batteries | Polymer | Molecular separations | Interfacial transport | Adsorption and carbon capture
Methods: Molecular dynamics | Monte Carlo | Density functional theory | Machine learning | Statistical mechanics | Mean-field and lattice models

EDUCATION

PhD in Chemical Engineering — Oct 2020 – Sept 2024
Department of Chemical Engineering, University of Manchester, UK

MSc in Advanced Chemical Engineering — Sept 2016 – Nov 2017
Department of Chemical Engineering, Imperial College London, UK

BEng Hons. in Chemical Engineering with Year Long Work Placement — Sept 2012 – Jun 2016
Department of Chemical Engineering, University of Bath, UK

ACADEMIC APPOINTMENTS & RESEARCH EXPERIENCE

Research Associate (Postdoctoral) — Dec 2024 – present
Chemistry, School of Natural and Environmental Sciences
Chemical Engineering, School of Engineering, Newcastle University, UK
Research visits undertaken during this appointment:
- **Short-term Visiting Scholar**, Eastern Institute of Technology, Ningbo, China — Dec 2025 – Jan 2026
- **Visiting Scholar**, Department of Chemical Engineering, University of Manchester, UK — Dec 2024 – Oct 2025

Research Assistant — Sept 2024 – Nov 2024
Department of Chemical Engineering, University of Manchester, UK

Research Assistant — Jan 2018 – Dec 2019
Department of Chemical and Biomolecular Engineering, Clemson University, US

Special Research Student — Jun 2017 – Sept 2017
Graduate School of Frontier Sciences, University of Tokyo, Japan
International research placement during MSc, Imperial College London

Development Technician, Freudenberg Performance Materials, UK — Jul 2014 – Jul 2015
Industrial placement year during BEng, University of Bath

RESEARCH FUNDING & COMPETITIVE SUPPORT

USD 5,000	**Google Cloud Research Credits** | Google Cloud	2026
GBP 1,500	**Researcher Development and Travel Grant** (2023, 2024) **Researcher Development Grant** (2022) | Royal Society of Chemistry	2022, 2023, 2024

HONOURS & AWARDS

Chinese Government Award for Outstanding Self-financed Students Abroad | China Scholarship Council — 2023
World Association of Membrane Societies Oral Presentation Award | 13th International Congress on Membranes and Membrane Processes — 2023
Graduate Teaching Assistant Award | University of Manchester — 2021
Overseas Research Scholar Award | Four-year PhD scholarship | University of Manchester — 2020-2024

ACADEMIC LEADERSHIP & SERVICE

Committee Member | Statistical Mechanics & Thermodynamics Group, Royal Society of Chemistry — 2025–
Session Chair | Modelling and Simulation in Membrane Science, Euromembrane 2024 — 2024

PUBLICATIONS

7 peer-reviewed papers | 5 first-author | 3 corresponding-author. *Corresponding author.

Peer-reviewed Articles

7. Mai, Z., Yoshioka, T., Deshmukh, A., **Yuan, T.**, Zhu, J., Yuan, J., Gonzales, R. R., Yamamoto, A., Shi, Y., Fu, W., Guan, K., Li, Z., Zhang, P., Lienhard, J. H., *Matsuyama, H., "Dynamic Sub-Nanoscale Water Fingers in Interfacial Polymerization", ***Small*** (2025), 2504497.
6. Yang, X., Feng, Z., Alshurafa, M., Yu, M., Foster, A. B., Zhai, H., **Yuan, T.**, Xiao, Y., D'Agostino, C., Ai, L., Perez-Page, M., Smith, K., Foglia, F., Lovett, A., Miller, T. S., *Chen, J., *Budd, P. M., *Holmes, S. M., "Durable Proton Exchange Membrane Based on Polymers of Intrinsic Microporosity for Fuel Cells", ***Advanced Materials*** 37 (2025), 2419534.
5. ***Yuan, T.**, Sarkisov, L., "How 2D nanoflakes improve transport in mixed matrix membranes: insights from a simple lattice model and dynamic mean field theory", ***ACS Applied Materials & Interfaces*** 16 (2024), 8184–8195.
4. ***Yuan, T.**, De Angelis, M. G., Sarkisov, L., "Simple lattice model explains equilibrium separation phenomena in glassy polymers", ***Journal of Chemical Physics*** 159 (2023).
3. **Yuan, T.**, DeFever, R. S., Zhou, J., Cortes-Morales, E. C., *Sarupria, S., "RSeeds: Rigid seeding method for studying heterogeneous crystal nucleation", ***Journal of Physical Chemistry B*** 127 (2023), 4112–4125.
2. ***Yuan, T.**, Sarkisov, L., "Lattice Model of Fluid Transport in Mixed Matrix Membranes", ***Advanced Theory and Simulations*** 5 (2022), 2200159.
1. **Yuan, T.**, Farmahini, A. H., *Sarkisov, L., "Application of Dynamic Lattice Mean Field Theory to fluid transport in slit pores", ***Journal of Chemical Physics*** 155 (2021), 074702.

Preprints and Manuscripts Under Review

2. Sana, B., Power, S. D. T., **Yuan, T.**, Jennings, J., Dawson, J., Knight, J. G., *Doherty, S., *Mamlouk, M., "Structure–property relationship of dialkyl and spirocyclic piperidinium-based poly(biphenyl piperidinium) anion exchange membranes". *Under review*.
1. ***Yuan, T.**, Giro, R., Steiner, M. B., Hsu, H., Sarkisov, L., "Emergence of the Robeson bound in non-equilibrium molecular dynamics simulations", ***ChemRxiv*** (2024).

CONFERENCE PRESENTATIONS

Oral Presentations

4. "Effect of electric fields on ion transport at the argyrodite Li_6PS_5Cl–Li metal interface", *Solid State Ionics 25*, Singapore, July 2026.
3. "Gas separation through polymer membranes: from both equilibrium and non-equilibrium molecular simulations", *Euromembrane 2024*, Prague, Czechia, Sept 2024.
2. "Equilibrium separation phenomena in a simple lattice model of polymer membrane", *13th International Congress on Membranes and Membrane Processes*, Chiba, Japan, July 2023. **Oral Presentation Award**.
1. "Lattice Model of Fluid Transport in Mixed Matrix Membranes", *The 27th Thermodynamics Conference*, Bath, UK, Sept 2022.

Posters

5. "Effect of electric fields on ion transport at the argyrodite Li_6PS_5Cl–Li metal interface", *Faraday Institution Conference 2026*, Nottingham, UK, Sept 2026.
4. "Atomistic exploration of argyrodite–Li metal interfaces for solid-state batteries", *Faraday Institution Conference 2025*, Warwick, UK, Sept 2025.
3. "How 2D nanoflakes improve transport in mixed matrix membranes: insights from a simple lattice model and dynamic mean field theory", *13th International Congress on Membranes and Membrane Processes*, Chiba, Japan, July 2023.
2. "Application of dynamic mean field theory to study fluid transport in membranes", *The 14th International Conference on Fundamentals of Adsorption*, Denver, CO, US, May 2022.
1. "Rigid Seeding: A Computationally Efficient Method to Study Heterogeneous Nucleation in Molecular Simulations", *Crystal Growth and Assembly Gordon Research Conference*, Manchester, NH, US, June 2019.

TEACHING & EDUCATIONAL RECOGNITION

Graduate Teaching Assistant, University of Manchester, UK — Oct 2020 – Sept 2024

Postgraduate Advanced Gas Separations | Chemical Engineering Molecular Simulations | Research Techniques and Methods | Reaction System Design

Undergraduate Batch Processing | Catalytic Reactor Engineering | Chemical Engineering Optimisation | Process Control | Process Design

Graduate Teaching Assistant, Clemson University, US		Aug 2018 – Jan 2019
Undergraduate	Thermodynamics II	

PROFESSIONAL RECOGNITION & MEMBERSHIPS

Member	**MRSC** | Royal Society of Chemistry (RSC)	2021–
Associate Fellow	**AFHEA** | Advance HE (formerly the Higher Education Academy)	2024–
Associate Member	**AMIChemE** | Institution of Chemical Engineers (IChemE)	2024–2025

COMPUTATIONAL & TECHNICAL EXPERTISE

Simulation	LAMMPS | GROMACS | VASP | CP2K | DFTB+
Programming	Python | C++ | C | Fortran | Bash | Tcl | LaTeX | CUDA
HPC	Linux/Unix | HPC environments | GPU computing | MPI/OpenMP
Web	HTML | CSS | JavaScript
Additional	First Aid